Grace Guo, Ph.D.

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RESEARCH & TEACHING INTERESTS

My research seeks to develop human-centered visual analytics tools that make AI systems more interpretable, accountable, and trustworthy. My teaching interests includes visual analytics, my area of research expertise, as well as HCI and AI.

EDUCATION

2024 **Ph.D. in Human-centered Computing,** Georgia Institute of Technology, Atlanta, GA
Faculty Advisor: Alex Endert, Ph.D.
Funding: IBM Ph.D. Fellowship (1 of 10 worldwide), 2023

2018 **B.S. in Human-Computer Interaction and Cognitive Science,** Carnegie Mellon University, Pittsburgh, PA

RESEARCH EXPERIENCE

2024- **Postdoctoral Fellow,** John A. Paulson School of Engineering & Applied Sciences (SEAS), Harvard University, Cambridge, MA

PUBLICATIONS

Journal Articles

[1] Salma Abdel Magid, **Grace Guo**, Esin Tureci, Amaya Dharmasiri, Vikram V. Ramaswamy, Hanspeter Pfister, Olga Russakovsky. 2026. Bias at the End of the Score. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).*

[2] Salma Abdel Magid, Weiwei Pan, Simon Warchol, **Grace Guo**, Junsik Kim, Mahia Rahman, and Hanspeter Pfister. 2025. Is What You Ask For What You Get? Investigating Concept Associations in Text-to-Image Models. *Transactions on Machine Learning Research (TMLR).*

[3] Simon Warchol, **Grace Guo**, Johannes Knittel, Dan Freeman, Usha Bhalla, Jeremy L Muhlich, Peter K Sorger, and Hanspeter Pfister. 2025. SEAL: Spatially-resolved Embedding Analysis with Linked Imaging Data. *IEEE Transactions on Visualization and Computer Graphics (VIS).*

[4] **Grace Guo**, Subhajit Das, Jian Zhao, and Alex Endert. 2025. More Like Vis, Less Like Vis: Comparing Interactions for Integrating User Preferences Into Partial Specification Recommenders. *IEEE Transactions on Visualization and Computer Graphics (TVCG).*

[5] Mark S Keller, Eric Mörth, Thomas C Smits, Simon Warchol, **Grace Guo**, Qianwen Wang, Robert Krueger, Hanspeter Pfister, and Nils Gehlenborg. 2025. The State of Single-Cell Atlas Data Visualization in the Biological Literature. *IEEE Computer Graphics and Applications.*

[6] **Grace Guo**, Lifu Deng, Animesh Tandon, Alex Endert, and Bum Chul Kwon. 2024. MiMICRI: Towards Domain-centered Counterfactual Explanations of Cardiovascular Image Classification Models. *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT).* 1–14.

[7] **Grace Guo**, Aishwarya Mudgal Sunil Kumar, Adit Gupta, Adam Coscia, Chris MacLellan, and Alex Endert. 2024. Visualizing Intelligent Tutor Interactions for Responsive Pedagogy. *Proceedings of the 2024 International Conference on Advanced Visual Interfaces (AVI).* 1–9.

[8] **Grace Guo**, John Stasko, and Alex Endert. 2024. What We Augment When We Augment Visualizations: A Design Elicitation Study of How We Visually Express Data Relationships. *Proceedings of the 2024 International Conference on Advanced Visual Interfaces (AVI).* 1–6.

[9] Anh-Ton Tran, **Grace Guo**, Jordan Taylor, Katsuki Chan, Elora Raymond, and Carl DiSalvo. 2024. Situating Data Sets: Making Public Data Actionable for Housing Justice. *Proceedings of the 2024 CHI conference on human factors in computing systems (CHI).* 1–16.

[10] **Grace Guo**, Ehud Karavani, Alex Endert, and Bum Chul Kwon. 2023. Causalvis: Visualizations for Causal Inference. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI).* 1–20.

[11] **Grace Guo**, Maria Glenski, ZhuanYi Shaw, Emily Saldanha, Alex Endert, Svitlana Volkova, and Dustin Arendt. 2021. Vaine: Visualization and AI for natural experiments. *Proceedings of the 2021 IEEE Visualization Conference (VIS).* IEEE, 21–25.

[12] Fabian Sperrle, Mennatallah El-Assady, **Grace Guo**, Rita Borgo, D Horng Chau, Alex Endert, and Daniel Keim. 2021. A Survey of Human-Centered Evaluations in Human-Centered Machine Learning. *Computer Graphics Forum*, Vol. 40. Wiley Online Library, 543–568.

[13] Austin P Wright, Zijie J Wang, Haekyu Park, **Grace Guo**, Fabian Sperrle, Mennatallah El-Assady, Alex Endert, Daniel Keim, and Duen Horng Chau. 2020. A comparative analysis of industry human-AI interaction guidelines. *arXiv preprint arXiv:2010.11761* (2020).

[14] **Grace Guo**, Bianchi Dy, Nazim Ibrahim, Sam Conrad Joyce, and Ate Poorthuis. 2020. Examining Design-Centric Test Participants in Graphical Perception Experiments. *EuroVis (Short Papers).* 43–47.

[15] Ate Poorthuis, Lucas van der Zee, **Grace Guo**, Jo Hsi Keong, and Bianchi Dy. 2020. Florence: a Web-based Grammar of Graphics for Making Maps and Learning Cartography. *Cartographic Perspectives* 96 (2020), 32–50

[16] Sam C Joyce, **Grace Guo**, Bianchi Dy, Nazim Ibrahim, and Ate Poorthuis. 2019. Seeing numbers: Considering the effect of presentation of engineering data in design. In Proceedings of IASS Annual Symposia, Vol. 2019. *International Association for Shell and Spatial Structures* (IASS), 1–8.

PRESENTATIONS

MiMICRI: Towards Domain-centered Counterfactual Explanations of Cardiovascular Image Classification Models, Explainable AI, ACM FAccT, 2024

Visualizing Intelligent Tutor Interactions for Responsive Pedagogy, Visual Tools for Education, ACM AVI, 2024

When We Augment Visualizations: A Design Elicitation Study of How We Visually Express Data Relationships, Visualization II, ACM AVI, 2024

Situating Data Sets: Making Public Eviction Data Actionable for Housing Justice, Politics of Datasets, ACM CHI, 2024

Visualizations in Context: Toolkits for Expressive Visualization Augmentation, Invited Talk @ Tableau Research, Salesforce, 2024

Causalvis: Visualizations for Causal Inference, Making Sense & Decisions with Visualization, ACM CHI, 2023

Flexible and Expressive Augmentation of Domain-Specific Visualizations, Doctoral Colloquium, IEEE VIS, 2022

VAINE: Visualization and AI for Natural Experiments, AI+VIS, IEEE VIS, 2021

Survey of Evaluations in Human-Centered Machine Learning, State-of-the-Art Reports (STARs), EuroVis, 2021

TEACHING EXPERIENCE

Teaching Assistant

Responsibilities included preparing and giving guest lectures, creating and updating programming homework, and guiding students on their research projects.

- Harvard University
  - Visualization, 90 undergraduate students, SEAS (Fall 2025)
  - Data Visualization for Analysis and Communication, 20 graduate students, Harvard Business School (Spring 2025)
- Georgia Institute of Technology
  - Introduction to Information Visualization, 100 undergraduate students (Spring 2023)
  - Issues in Human-Centered Computing, 20 graduate students (Spring 2022)
  - Computing, Society and Professionalism, 20 undergraduate students (Summer 2021)
  - Information Visualization, 100 graduate students (Fall 2020)
- Carnegie Mellon University
  - Cognitive Psychology, Undergraduate Course (Fall 2016, Fall 2017)
  - Fundamentals of Programming and CS, Undergraduate course (Fall 2015, Spring 2016)

PROFESSIONAL SERVICE

Committee Work

Member, Conference Organizing Committee IEEE Vis Papers 2026
Member, Conference Organizing Committee Information+, IEEE Vis Posters Track 2025

Reviewing

CHI Papers, VIS Full Papers, EuroVis Full Papers, CG&A Special Issue, TVCG Journal Papers, 2023, 2024, 2025, 2026

ADDITIONAL PROFESSIONAL EXPERIENCE

Research Intern, IBM Research, Cambridge, MA (Summer 2022 and Summer 2023)
Research Intern, Pacific Northwest National Laboratory, Richland, WA (Summer 2020)